

Design and characteristics of a tapered amplifier diode system by seeding with continuous-wave and mode-locked external cavity diode laser

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Abstract. We present a detailed design of a simple, stable, and compact tapered amplifier (TA) diode system. Heat conductivity, strain release, and system alignment are taken into consideration for this design. Convenient access to the three-dimensional adjustments for collimation makes the system easy to align. The optical amplification is discussed as a function of injection of current and operating temperature for both a continuous-wave external cavity diode laser (CW-ECDL) and a mode-locked external cavity diode laser (ML-ECDL). With 5-mW CW-ECDL seeding power, 240 mW is achieved after the TA, 115 mW (48%) of which can be coupled into a single-mode fiber. With 0.75-mW average ML-ECDL seeding power, 80-mW average power, or about 11-W peak power (843-MHz repetition rate and 8-ps pulse width), is achieved after the TA, 39 mW (49%) of which can be coupled into a single-mode fiber. In both cases, the amplified light maintains the optical properties of the seeding light, showing the same mode suppression. The output power spatial mode quality is characterized with a camera and BeamView software. Output power stability is also discussed. © 2006 Society of Photo-Optical Instrumentation Engineers. [DOI: 10.1117/1.2404925]

Subject terms: continuous-wave external cavity diode laser; mode-locked external cavity diode laser; tapered amplifier diode system; antireflection coating; collimation; system alignment; amplification.

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1 Introduction

External cavity diode lasers (ECDL) are commonly used in many experiments. Low-cost diodes, narrow-linewidth, and wide-frequency tunability are the main advantages of ECDL systems. Frequency-selective feedback is typically achieved via a grating in either the Littrow¹⁻³ or Littman-Metcalf^{4,5} configurations. With either configuration, the output power of the ECDL is limited by the risk of the high circulating power inside the external cavity damaging the diode. The typical continuous-wave ECDL (CW-ECDL) output power is 5–20 mW. After beam-shaping, Faraday isolation, and fiber coupling, only 3–10 mW remain. The typical average output power of a mode-locked ECDL (ML-ECDL) is only 1–2 mW.⁶ A tapered amplifier (TA) diode can make the optical power much higher while maintaining all the optical properties of the seeding ECDL.

In the past, most optical amplifiers were either extremely expensive or difficult to make reliable. Recently, lower-cost tapered amplifier diodes have become commercially available, making diode optical amplifiers more practical. With improvements in antireflection (AR) coating technology, laser action of the tapered amplifier itself can be avoided by coating both the front and back facets, giving the laser system the best possible performance. In this paper, we inves-

tigate a TA diode system that can amplify either a CW-ECDL or an ML-ECDL. The TA diode is manufactured by the German company Eagleyard Photonics. The gain region of the TA diode is 2.75 mm long with 3- μ m and 190- μ m input and output apertures, respectively. According to the manufacturer's datasheet, the maximum output power is about 500 mW, within a wavelength range from 785 to 805 nm.

This paper is organized in the following manner. Section 2 gives the details of the TA diode system design and alignment. The experimental setup is discussed in Section 3, followed by the experimental results from this TA diode system in Section 4. Finally, we offer some concluding remarks in Section 5.

2 TA Diode System Design and Alignment

We designed a simple, stable, and compact TA diode system (Fig. 1) that consists of a TA diode, two aspheric collimation lenses ($f=4.5$ mm 0.55 NA, Thorlabs C230TM-B) threaded into a homemade tube inserted into a modified fiber holder (Newport F-91-C1-T), a Plano-Convex cylindrical lens ($f=50$ mm), and a $4\times3\times2.2$ -in. aluminum heat sink. The C-Mount TA diode has an antireflection coating on both ends and a 2.75-mm-long tapered gain region, which allows for a weak signal to be injected into the narrow end and the amplified light to come out from the broad

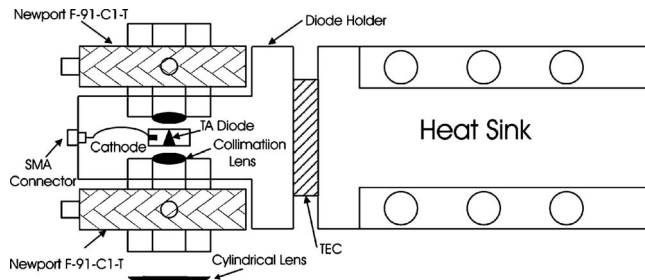


Fig. 1 Top view of the TA diode system. Heat sink and TEC are for heat conductivity; Newport F-91-C1-T and cylindrical lens are for collimation adjustment; SMA connector is for strain release.

end after traveling through the gain region. It is necessary to have a compact and stable mount for the TA diode and the collimators, because the input aperture of the TA diode is only $3\ \mu\text{m}$. The C-Mount TA diode is screwed onto a homemade brass diode holder. The modified commercially available fiber holders with collimation lenses in them are attached on both sides of the diode holder. These fiber holders allow adjustments in three dimensions, ensuring good coupling of the input light and collimation of the output light. In earlier work,⁷ the z -direction of the output collimator was not accessible from the outside. However, in our design, all these adjustments are accessible from the outside of the housing surrounding the TA diode mount.

Thermal variations can cause the output power of the TA to fluctuate. To thermally control this system, we attached the brass diode holder mount to an aluminum heat sink with a 2×2 -in. thermoelectric cooler in between. In addition, the aluminum housing that surrounds the TA is designed to minimize the amount of air around the TA to help reduce any thermal variations that air movement could cause. Water cooling was not needed or used in our design.

The soldered connection of the diode cathode tab on the C-Mount is very fragile and needs to be attached quickly in order to prevent damage to the TA diode by overheating. Therefore, in addition to screwing the TA diode onto the homemade brass diode holder, we also soldered the cathode tab to a wire from an SMA connector attached to the front case of TA diode system in order to serve as a strain release. This helps prevent accidental damage to this fragile connection.

Good alignment plays a big role in the amplification performance. In this system, we first turn on the TA without any seeding light; because of the AR coatings, the amplified spontaneous emission (ASE) power is only from 0.8 to 3 mW depending on the TA drive current. On the input side, the three-dimensional adjustments of the input lens allow the emitted ASE to be collimated. The ASE beam profile on the input side is roughly circular and can therefore be fiber coupled with about 45–55% efficiency into a single-mode fiber. The manufacture datasheet lists the perpendicular divergent external half-angle as 14° and the parallel divergent external half-angle as 5° . A 4.5-mm spherical lens was used to collimate the perpendicular divergent beam. Using geometric optics, we calculated a perpendicular collimated beam size of 2.242 mm and the location where the parallel beam focuses. Knowing these two numbers, we determined that a 54-mm focal length cylindrical

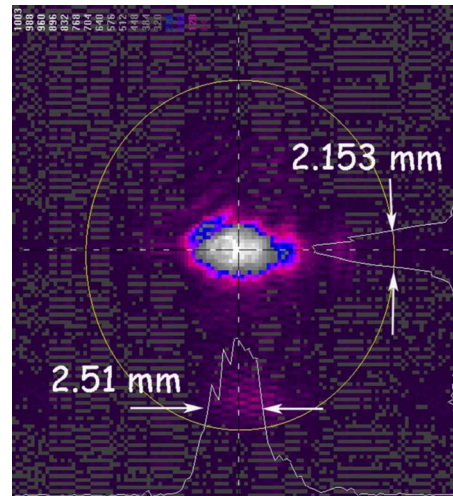


Fig. 2 Beam shape after TA. The beam viewer software shows that the parallel and perpendicular widths are 2.51 mm and 2.153 mm. The beam shape is not perfectly circular, but good enough to give 45–55% coupling efficiency to a single-mode fiber.

lens would be needed to collimate the parallel divergent beam to the same size as the collimated perpendicular beam. We used the closest commercially available cylindrical lens, which has a 50-mm focal length. In this way, the beam shape will be close to circular before going through the Faraday isolator and being fiber-coupled. Figure 2 gives details about the beam shape after the TA diode and both lenses. The parallel and perpendicular beam widths were measured with BeamView software to be 2.51 mm and 2.153 mm, respectively, very close to the 2.242 mm calculated earlier. The ratio of parallel to perpendicular is about 5:4.3; not a perfectly circular beam, but close enough to give 45–55% fiber coupling efficiency into a single-mode fiber. Improvements to the circularity of the beam may be possible with changing the focal length of the cylindrical lens. The whole alignment process is quite simple because of the easy access to the three-dimensional adjustments on both collimation lenses.

3 Experimental Setup

The ECDL used in the experiment (Fig. 3) is a 17.8-cm Littrow cavity corresponding to the fundamental free spectral range of 843 MHz. Both DC and AC current supplies are connected to the ECDL via a bias tee. A 5-dBm AC sine

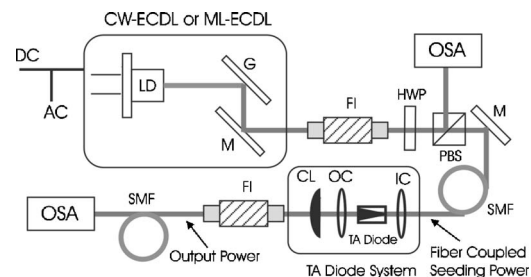


Fig. 3 Experimental setup. G: grating; LD: laser diode; M: mirror; FI: Faraday isolator; HW: half-wave plate; PBS: polarized beamsplitter; SMF: single-mode fiber; IC: input collimator; OC: output collimator; CL: cylindrical lens; OSA: optical spectrum analyzer.

wave modulation at 843 MHz is produced by a signal generator (HP 8648A) and is amplified to 21 dBm (Mini-Circuits ZQL-900LNW) before being combined with the DC current and applied to the ECDL. With only the DC current on, it behaves as a CW-ECDL; while with the sinusoidal AC current at 843 MHz in addition to the DC current, it acts as a ML-ECDL. Output light from the CW-ECDL (or ML-ECDL) goes through a Faraday isolator and then is coupled into a single-mode fiber, the output of which is the seeding power for the TA diode system. ILX Lightwave LDC-3744B diode current controller is used as the current source for the TA diode. This instrument provides a maximum DC current of 4 A. The amplified light after another Faraday isolator is considered the output power from TA diode system.

4 Characteristics of the Tapered Amplifier Diode System

With CW-ECDL seeding, Fig. 4 shows plots of the TA output power as a function of drive current for two different temperatures, 18°C and 22°C. Since both the TA diode and the collimation lens are mounted to the same piece, they will tend to have the same thermal expansion rate. Thus, we expect that even with a 4°C change in temperature, only a small misalignment will occur. However, we did readjust the fiber coupler for the seeding power after every temperature adjustment to maximize the coupling efficiency of the seeding power into the TA diode and found that very little correction was needed. Both plots show that for a higher TA drive current, higher amplification is achieved. By comparing the amplification for the same TA drive current at different temperatures, it can be seen that by lowering the temperature from 22°C to 18°C, the amplification increases by roughly 10–20%. With 5-mW seeding power at 18°C, 240 mW is achieved after TA, and 115 mW (48%) can be coupled into a single-mode fiber.⁷

Figure 5 shows the TA output power versus drive current at 18°C after seeding with the ML-ECDL. With 0.75-mW seeding power, 80-mW average power can be achieved by driving the TA at 1.75 Amps at 18°C, 39 mW (49%) of which can be coupled into a single-mode fiber.^{6,8} A typical interferometric autocorrelation tracer⁹ shown in Fig. 6, which displays a pulse with a full-width at half-maximum (FWHM) measurement of 12 ps in duration. Deconvolving this measurement with a Gaussian pulse shape yields a temporal pulse width of 8.5 ps. However, in many cases, the mode-locked pulses obtained from an actively mode-locked semiconductor laser are asymmetric in shape, with fast rise times and slower fall times.^{10–12} A more realistic situation is to deconvolve the measured pulse shapes with a single side exponential. This implies mode-locked pulses with a temporal width closer to 6 ps. Taking a conservative approach, pulse widths of 8 ps are assumed in this paper. With a repetition rate of 843 MHz (or 1.19 ns), the 80-mW average mode-locked power corresponds to a peak power of about 11 W. The 15- μ m actuator step size of our autocorrelator is not fine enough to resolve the details of the electric field modulation under the fitting curve, but it still shows the field modulation properties.

Figure 7 shows the optical spectrum of the seeding light and the light after amplification. In both the CW and ML cases, the amplified light keeps the same optical spectrum

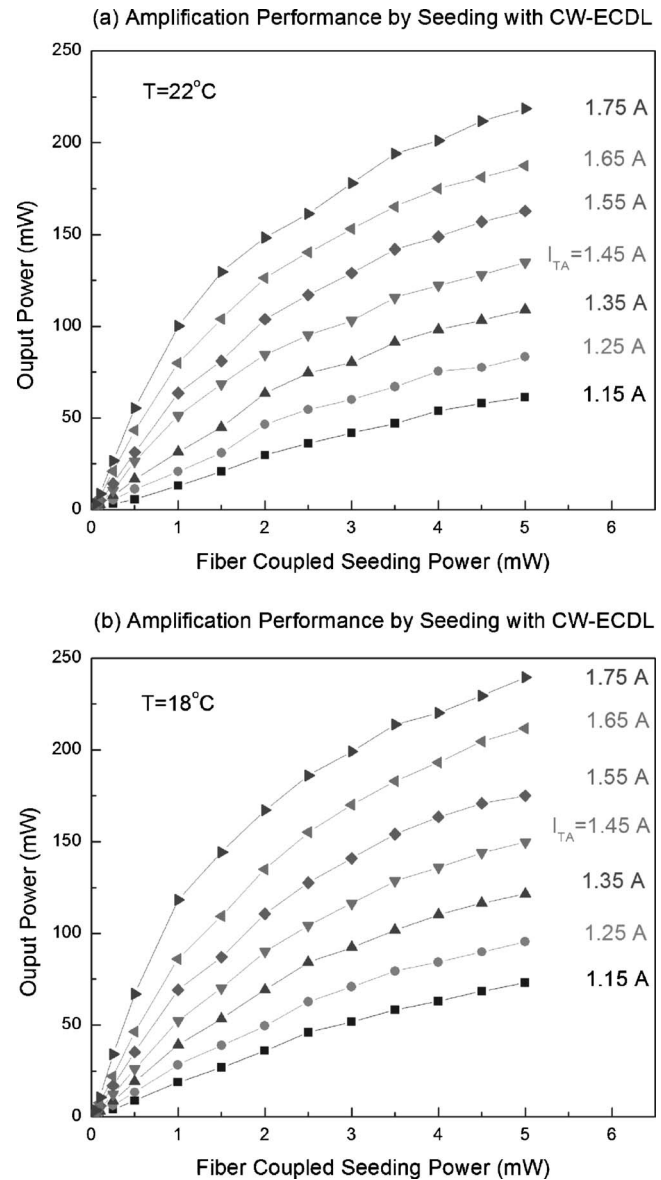


Fig. 4 Temperature and current dependence of TA amplification by seeding with CW-ECDL. Both plots show the amplification when the TA is operated at seven different currents. (a) $T=22^{\circ}\text{C}$; (b) $T=18^{\circ}\text{C}$. Lower temperature and higher drive current give higher amplification.

as the seeding light. Since the TA diode is AR coated at both ends, with a drive current from 1.15 A to 1.75 A, the ASE varies from 0.8 mW to 3 mW and the optical spectrum is flat over about 20 nm. By comparing ASE with the output power in Fig. 4, we can tell that except when the seeding power is small, the ASE is negligible compared to the amplified light. After being amplified by the TA, the mode suppression is roughly 38 dB for CW seeding and 32 dB for ML seeding, which equal the mode suppression of the seeding CW-ECDL and the ML-ECDL. The spectral width of amplified ML-ECDL is 0.16 nm, roughly corresponding to 72 longitudinal modes, which also maintains the spectral width of the seeding ML-ECDL. The fraction of ASE background in the TA output is obtained by integrating the power spectral density on a linear scale in Fig.

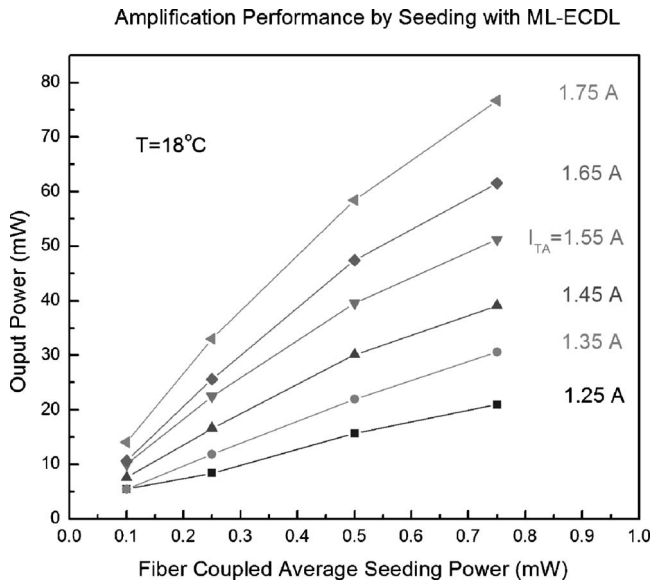


Fig. 5 Amplification performance by seeding with ML-ECDL at different TA currents.

4(b), yielding 1.65% for CW seeding and 2.92% for ML seeding at 22°C. We can see obvious improvements of our system when compared to the results of previous experiments,⁷ which achieved, with CW seeding, 5.6% at 16°C and 1.4% at 5°C. The improvement in our case is probably due to a better AR coating on our TA diode. In general, thermal instabilities make the output power fluctuate; however, with the considerations mentioned earlier, the output power after this TA diode system is very stable once the system reaches thermal equilibrium. As shown in Fig. 8, with 2-mW CW seeding power at 22°C, during the first couple of minutes after turning on the TA, the power

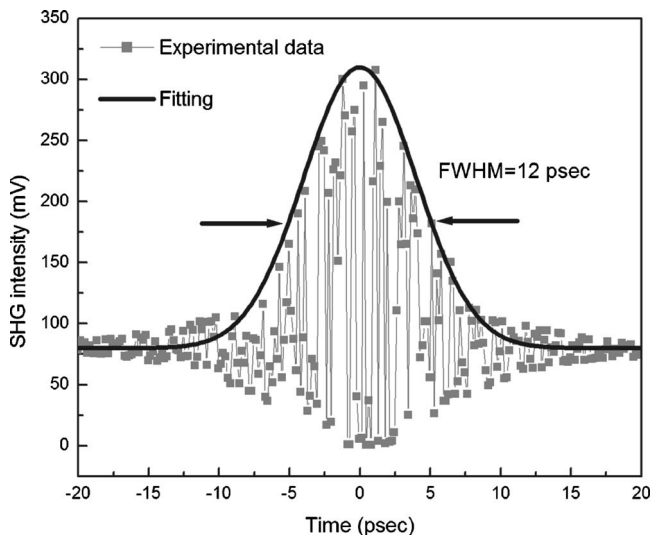


Fig. 6 Interferometric autocorrelation trace of the amplified pulses. Autocorrelation of the ML laser pulse shows a FWHM of 12 ps. This corresponds to a pulse duration of approximately 9 ps by taking the deconvolution between pulse shapes. Since the stepping of the autocorrelator is not fine enough, the structure beneath the fitting curve does not represent the details of the electric field modulation, but still shows field modulation property.

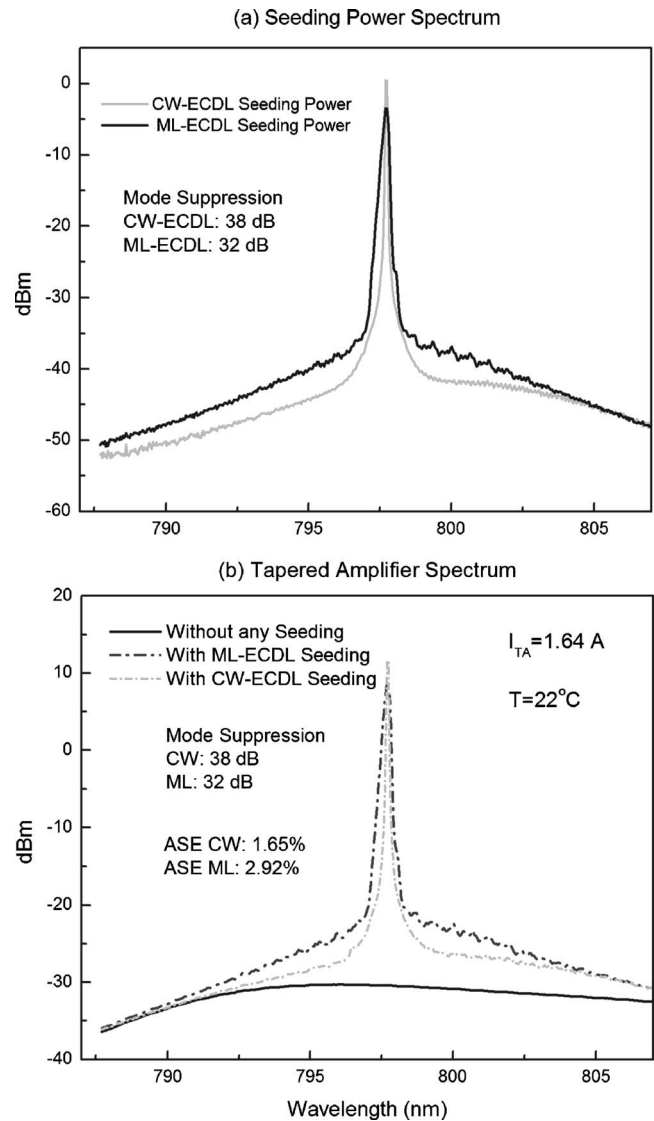


Fig. 7 TA optical spectrum with ML and CW seeding. Mode suppression is about 32 dB and 38 dB, respectively.

reaches 160 mW and then drops immediately due to sudden temperature changes associated with turning on the TA diode. After about 15 minutes, when the system reaches the thermal equilibrium, the output power remains at 140 mW for more than the 50 minutes measured.

5 Summary

We have shown a simple and stable design for a tapered amplifier diode system, taking into consideration the thermal management of the heat dissipation, three-dimensional collimation adjustments, and strain release. The easily accessible three-dimensional collimation adjustments allow us to precisely align the input and output with the TA diode, giving better output amplification performance. Both CW-ECDL and ML-ECDL were constructed and used to inject optical power into the TA diode system. CW power after amplification reached 240 mW with 5-mW seeding power. This can be used as a tunable laser source for a Lidar project, like water vapor profiles in the atmosphere¹³ or any other system that requires high power while maintaining

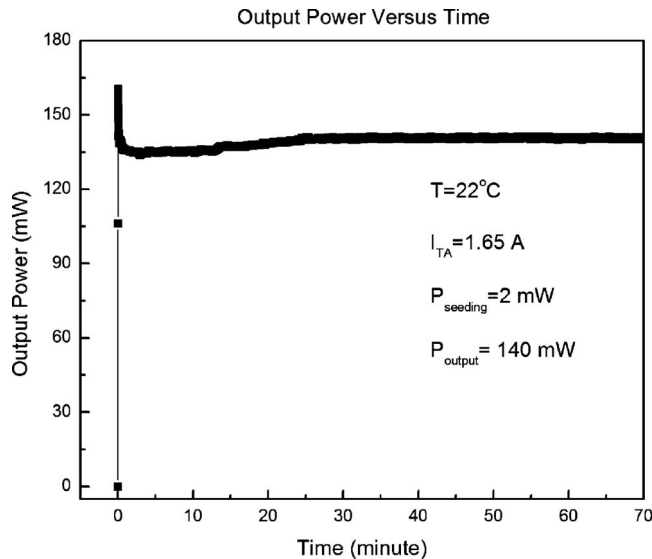
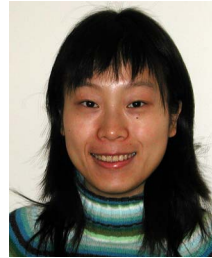


Fig. 8 TA output power versus time. TA diode system reaches thermal equilibrium 20 minutes after turning on the TA diode, and the output power remains 140 mW for more than 50 minutes.

the narrow linewidth and frequency tunability. With 0.75-mW average seeding power, ML average powers after amplification exceeded 80 mW with 8-ps pulse duration, translating to a peak power of around 11 W. These power levels should be sufficient to investigate nonlinear optical phenomena, such as self-phase modulation in a fiber or the Raman effect in a high-finesse cavity with hydrogen. We also showed the optical spectrum after the TA maintains the same characteristics as the seeding light and that the output spatial mode quality allowed 45–55% coupling efficiency into a single-mode fiber.

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